



High-Speed Single-Port Synchronous SRAM

RA1SH6144x32_8 6144X32, Mux 16, Drive 12

Features

- Precise Optimization for TSMC's Six-Layer Metal 0.18μm CMOS Process
- High Density (area is 1.173mm²)
- Fast Access Time (1.41ns at typical process, 1.80V, 25°C)
- Fast Cycle Time (1.36ns at typical process, 1.80V, 25°C)
- One Read/Write Port
- Configurable Word-Write Mask
- Completely Static Operation
- Near-Zero Hold Time (Data, Address, and Control Inputs)

Memory Description

The 6144X32 SRAM is a high-performance, synchronous single-port, 6144-word by 32-bit memory designed to take full advantage of TSMC's six-layer metal, 0.18μm CMOS process.

The SRAM's storage array is composed of six-transistor cells with fully static memory circuitry. The SRAM operates at a voltage of 1.8V ± 10% and a junction temperature range of 0°C to +125°C.

Pin Description

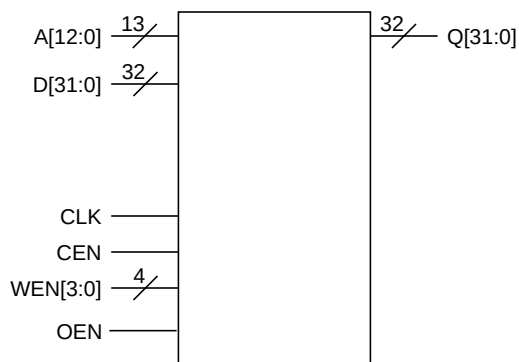
Pin	Description
A[12:0]	Addresses (A[0] = LSB)
D[31:0]	Data Inputs (D[0] = LSB)
CLK	Clock Input
CEN	Chip Enable
WEN[3:0]	Write Enable
OEN	Output Enable
Q[31:0]	Data Outputs (Q[0] = LSB)

Area

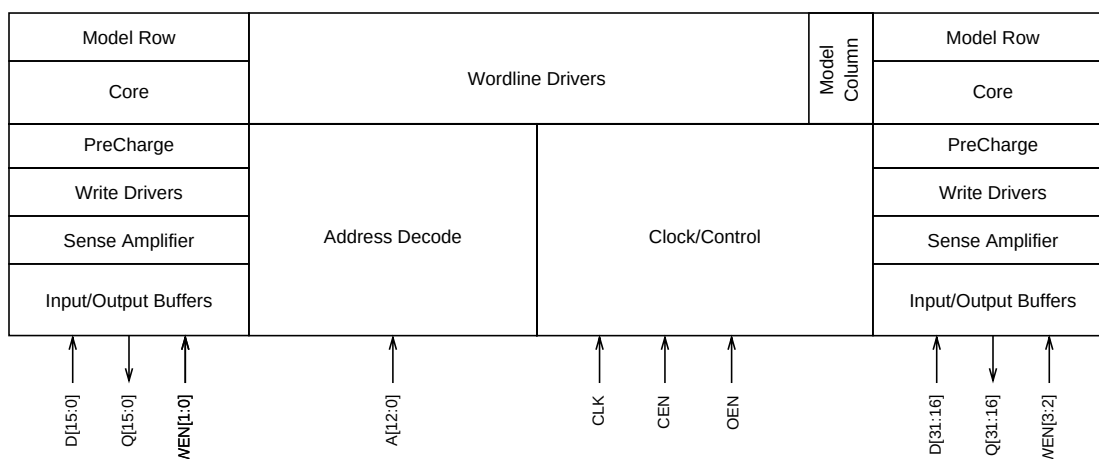
Area Type	Width (mm)	Height (mm)	Area (mm ²)
Core	1.080	1.086	1.173
Footprint	1.118	1.124	1.257

The footprint area includes the core area and user-defined power ring and pin spacing areas.

Symbol

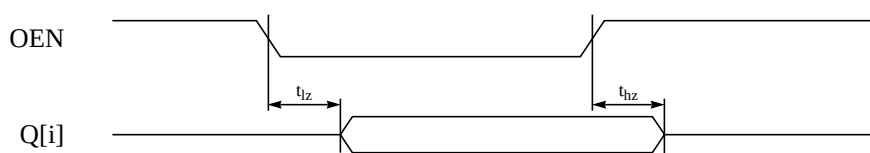


SRAM Block Diagram



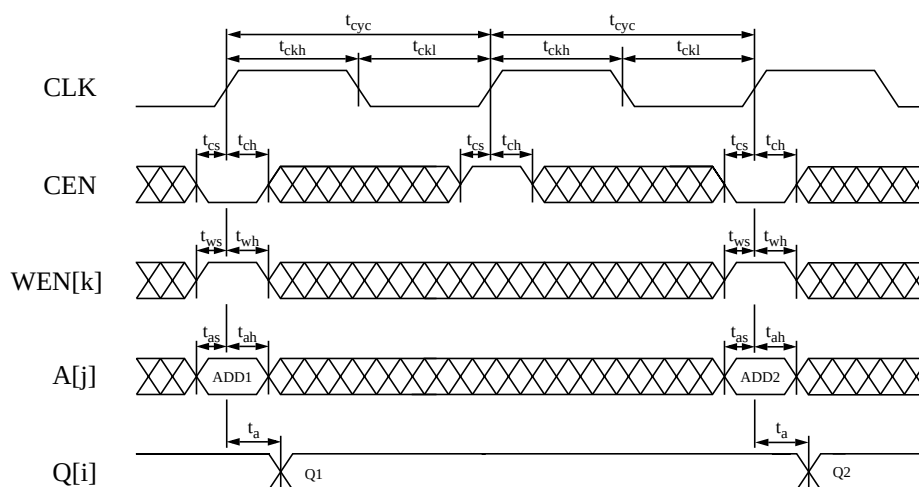
Mission Mode

Figure 1. Synchronous Single-Port SRAM Output-Enable Timing



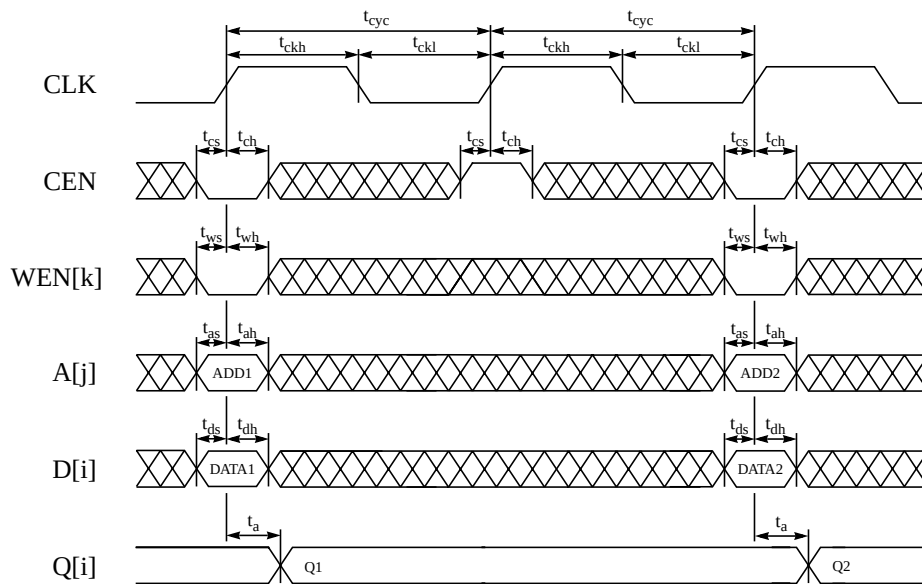
Rising signals are measured at 50% of VDD and falling signals are measured at 50% of VDD.
 Rising and falling slews are measured from 10% VDD to 90% VDD.

Figure 2. Synchronous Single-Port SRAM Read-Cycle Timing, with Word-Write Mask



Rising signals are measured at 50% of VDD and falling signals are measured at 50% of VDD.
 Rising and falling slews are measured from 10% VDD to 90% VDD.

Figure 3. Synchronous Single-Port SRAM Write-Cycle Timing, with Word-Write Mask



Rising signals are measured at 50% of VDD and falling signals are measured at 50% of VDD.

Rising and falling slews are measured from 10% VDD to 90% VDD.

SRAM Logic Table

CEN	WEN[k]	OEN	Data Out	Mode	Function
H	X	L	Last Data	Standby	Address inputs are disabled; data stored in the memory is retained, but the memory cannot be accessed for new reads or writes. Data outputs remain stable.
L	L	L	Data In	Write	The corresponding data partition is selected, and that data is written to the memory location specified on the address bus A[m-1:0], and driven through to the data output bus Q[n-1:0].
L	H	L	SRAM Data	Read	Data on the data output bus Q[n-1:0] is read from the memory location specified on the address bus A[m-1:0].
X	X	H	Z	High-Z	The data output bus Q[n-1:0] is placed in a high impedance state. Other memory operations are unaffected.

SRAM Timing: Mission Mode

Parameter	Symbol	Fast Process 1.98V, 0°C		Typical Process 1.80V, 25°C		Slow Process 1.62V, 125°C	
		Min (ns)	Max (ns)	Min (ns)	Max (ns)	Min (ns)	Max (ns)
Cycle time	t_{cyc}	0.95		1.36		2.26	
Access time ^{1,2}	t_a	0.93			1.41		2.32
Address setup	t_{as}	0.23		0.33		0.58	
Address hold	t_{ah}	0.06		0.07		0.10	
Chip enable setup	t_{cs}	0.27		0.36		0.56	
Chip enable hold	t_{ch}	0.00		0.00		0.00	
Write enable setup	t_{ws}	0.19		0.24		0.40	
Write enable hold	t_{wh}	0.00		0.00		0.00	
Data setup	t_{ds}	0.13		0.20		0.38	
Data hold	t_{dh}	0.00		0.00		0.00	
Output enable to hi-Z	t_{hz}		0.46		0.62		0.96
Output enable active ¹	t_{lz}		0.41		0.55		0.85
Clock high	t_{ckh}	0.08		0.11		0.18	
Clock low	t_{ckl}	0.11		0.17		0.28	
Clock rise slew	t_{ckr}		4.00		4.00		4.00
Output load factor (ns/pF)	K_{load}		0.27		0.36		0.53

¹ Parameters have a load dependence (K_{load}), which is used to calculate: $TotalDelay = FixedDelay + (K_{load} \times C_{load})$.

² Access time is defined as the slowest possible output transition for the typical and slow corners, and the fastest possible output transition for the fast corner.

Pin Capacitance

Pin	Fast Process 1.98V, 0°C	Typical Process 1.80V, 25°C	Slow Process 1.62V, 125°C
	Value (pF)	Value (pF)	Value (pF)
A[j]	0.053	0.052	0.052
D[i]	0.004	0.004	0.004
CLK	0.264	0.257	0.229
CEN	0.014	0.014	0.014
WEN[k]	0.120	0.117	0.117
OEN	0.010	0.010	0.010
Q[i]	0.019	0.019	0.018

Power

100.00MHz Operation

Condition	Fast Process 1.98V, 0°C	Typical Process 1.80V, 25°C	Slow Process 1.62V, 125°C
	Value (mA)	Value (mA)	Value (mA)
AC Current ¹	16.793	14.393	12.505
Read AC Current	14.765	12.584	10.902
Write AC Current	18.822	16.201	14.109
Peak Current	414.345	282.395	157.140
Deselected Current ^{2, 3}	0.000	0.000	0.000
Standby Current ⁴	leakage only	leakage only	leakage only

¹ Value assumes 50% read and write operations.² Value assumes all input and output signals are stable.³ If inputs toggle, current can be 30-40% of icc because the input latches are open, and internal logic can switch.⁴ Value assumes SRAM is deselected and only CLK switches.**Clock Noise Limit**

Signal	Fast Process 1.98V, 0°C		Typical Process 1.80V, 25°C		Slow Process 1.62V, 125°C	
	Pulse Width (ns)	Voltage (V)	Pulse Width (ns)	Voltage (V)	Pulse Width (ns)	Voltage (V)
CLK	10.00	0.82	10.00	0.85	10.00	0.83

The clock noise limit is the maximum CLK voltage allowable for the indicated pulse width without causing a spurious memory cycle or other memory failure.

Power and Ground Noise Limit

Signal	Fast Process 1.98V, 0°C	Typical Process 1.80V, 25°C	Slow Process 1.62V, 125°C
	Voltage (V)	Voltage (V)	Voltage (V)
Power	0.20	0.18	0.16
Ground	0.20	0.18	0.16

The power/ground noise limit is the maximum supply voltage transition allowable without causing a memory failure.